

Level 15 → Level 16

현재 비밀번호를 localhost의 30001번 포트에 SSL/TLS 암호화를 사용하여 전송하면 비밀번호를 얻을 수 있다고 한다. openssl s_client 명령어로 SSL/TLS 암호화를 사용했다. 그냥 이용하니 연결이 끊겨 -ign_eof 명령어를 붙이니 제대로 비밀번호가 나왔다.

```
bandit15@bandit:~$ echo "8xCjnmgoKbGLhHFAZlGE5Tmu4M2tKJQo" | openssl s_client -connect localhost:30001 -ign_eof
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
---
Certificate chain
 0 s:CN = SnakeOil
  i:CN = SnakeOil
  a:PKEY: rsaEncryption, 4096 (bit); sigalg: RSA-SHA256
  v:NotBefore: Jun 10 03:59:50 2024 GMT; NotAfter: Jun  8 03:59:50 2034 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIFBzCCAu+gAwIBAgIUaBLz7DBxA0IfojaL/WaJzE6Sbz7cwDQYJKoZIhvcNAQEL
BQAwEzERMA8GA1UEAwYIU25ha2VPaWwwHhcNMjQwNjEwMDM1OTUwWhcNMzQwNjA4
MDM1OTUwWjATMREwDwYDVQDDAHTbmFrZU9pbDCCAiIwDQYJKoZIhvcNAQEBBQAD
ggIPADCCAgCGgIBANI+P5QXm9Bj21FIPsQqbqZRb5XmSZZJYaam7EIJ16Fxedf+
jXAv4d/FVqiEM4BuSNsNmBmX2Gq0LAfN33h+RMTjRoMb8yBsZsC063MLfXCk4p+
09gtGP7BS6Iy5XdmfY/fPHvA3JDEScdlDDmd6Lsbdwhv93Q8M6POV09sv4HuS4t/
jEjr+NhE+Bjr/wDbyy7GL71BP1WPZpQnRE40zoSrt5+bZVLv0DWUFWinB0fLaGRk
GmI0r5EU0Ud7HpYyoIQbiNlePGfPpHRKnmdXTTEZEoxeWWAaM1VhPGqfrB/Pnca+
vAJX7iB0b3kHimnfV0ScsG/YAUR94wSELeY+ULEWJaELVUntrJ5HeRDiTChiVQ++
wnnNjNbepaW6shopybUF3XXfhIb4NvwLWpvoKFXVtcVjLOuJF0snVvpE+MRT0wacy
tHtjZs7Ao7GYxDz6H8AdBLKJW67uQon37a4MI260ADFMS+2vEAbNSFP+f6ii5mrB
18cY64ZaF6oU8bjGK7BARdX56bRc3WFyUBIGWAFHEuB948BcshXY7baf5jjzPmgz
-----
-----BEGIN CERTIFICATE-----
MIIFBzCCAu+gAwIBAgIUaBLz7DBxA0IfojaL/WaJzE6Sbz7cwDQYJKoZIhvcNAQEL
BQAwEzERMA8GA1UEAwYIU25ha2VPaWwwHhcNMjQwNjEwMDM1OTUwWhcNMzQwNjA4
MDM1OTUwWjATMREwDwYDVQDDAHTbmFrZU9pbDCCAiIwDQYJKoZIhvcNAQEBBQAD
ggIPADCCAgCGgIBANI+P5QXm9Bj21FIPsQqbqZRb5XmSZZJYaam7EIJ16Fxedf+
jXAv4d/FVqiEM4BuSNsNmBmX2Gq0LAfN33h+RMTjRoMb8yBsZsC063MLfXCk4p+
09gtGP7BS6Iy5XdmfY/fPHvA3JDEScdlDDmd6Lsbdwhv93Q8M6POV09sv4HuS4t/
jEjr+NhE+Bjr/wDbyy7GL71BP1WPZpQnRE40zoSrt5+bZVLv0DWUFWinB0fLaGRk
GmI0r5EU0Ud7HpYyoIQbiNlePGfPpHRKnmdXTTEZEoxeWWAaM1VhPGqfrB/Pnca+
vAJX7iB0b3kHimnfV0ScsG/YAUR94wSELeY+ULEWJaELVUntrJ5HeRDiTChiVQ++
wnnNjNbepaW6shopybUF3XXfhIb4NvwLWpvoKFXVtcVjLOuJF0snVvpE+MRT0wacy
tHtjZs7Ao7GYxDz6H8AdBLKJW67uQon37a4MI260ADFMS+2vEAbNSFP+f6ii5mrB
18cY64ZaF6oU8bjGK7BARdX56bRc3WFyUBIGWAFHEuB948BcshXY7baf5jjzPmgz
-----
```

```
.....
Post-Handshake New Session Ticket arrived:
SSL-Session:
  Protocol : TLSv1.3
  Cipher : TLS_AES_256_GCM_SHA384
  Session-ID: 92EDBC46BC4F8541B0479CEA4B509306436499A59372B19D25239FAEDCC8B8DA
  Session-ID-ctx:
  Resumption PSK: FBB4DD79E761EDD1B1FAD255D99D2069C818DDECD750D07213A4C69E23B4C397E5E58D795E27EF5386A3E83CF0C327EB
  PSK identity: None
  PSK identity hint: None
  SRP username: None
  TLS session ticket lifetime hint: 300 (seconds)
  TLS session ticket:
0000 - ff 0b dc 88 bd cd ad 80-94 63 03 cb eb 57 77 9b .....c...Ww.
0010 - 60 9d 4b cb 5e 0f 40 c7-6e a6 f1 1e 4c ba 3e 25 `K.^.@.n...L.>%
0020 - a5 1d 59 d0 a7 65 b6 73-74 58 e3 d1 88 7b 8d 76 ..Y...e.stX...{.v
0030 - 4c f1 c7 a5 42 4d 73 e1-e6 0b bc ce 26 54 20 5b L...BMs.....&T [
0040 - 72 ac 53 16 be 21 f4 ce-f4 26 0d 6b 73 04 f1 9a r.S...!...&.ks...
0050 - 8c 08 7e 45 b7 77 be e6-d4 b8 a7 a5 69 e9 39 ad ..~E.w.....i.9.
0060 - b3 60 32 9f e2 07 41 3b-94 79 31 2a 05 c6 af 6b `2...A;.y1*...k
0070 - b5 10 c9 66 fc e0 9c a7-81 27 05 b7 6b 46 4a 8a ...f.....'.kFJ.
0080 - be 9a 1d 35 5b bb 4a 9b-03 d9 64 27 57 33 d6 13 ...5[.J....d'W3..
0090 - ed d7 09 a8 a8 e2 d7 7d-a1 cb 45 a3 45 0a 30 22 .....}.E.E.0"
00a0 - 2f cb 78 52 fa 5f e2 c6-f9 7b 89 19 15 34 26 f2 /.xR.....{...4&.
00b0 - 70 c6 a4 66 86 27 6a 90-39 9f 74 1d 92 a6 99 15 p..f.'j.9.t....
00c0 - 9b a7 99 a5 4f 6c fd 5d-8a db 60 7c d2 df 33 d0 ....0L.]...'|...3.
00d0 - 09 3d ea b2 c5 6f 0d 76-01 76 b7 af 05 50 4d 1d .=...o.v...PM.

Start Time: 1780153203
Timeout : 7200 (sec)
Verify return code: 18 (self-signed certificate)
Extended master secret: no
Max Early Data: 0
---
read R BLOCK
Correct!
kSkvUpMQ7lBYyCM4GbpVcT1BfWRy0Dx
```

Level 16 → Level 17

일단 localhost의 31000~32000 포트 중 열려있는 포트를 찾아야 한다. nmap 명령어를 이용해 확인하니 ssl 이면서 echo가 아닌 포트는 31790 포트로 확인됐다.

```
bandit16@bandit:~$ nmap -sV -p 31000-32000 localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-30 15:07 UTC
Stats: 0:00:25 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Stats: 0:01:04 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 80.00% done; ETC: 15:08 (0:00:16 remaining)
Stats: 0:02:22 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 80.00% done; ETC: 15:10 (0:00:36 remaining)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00020s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
31046/tcp  open  echo
31518/tcp  open  ssl/echo
31691/tcp  open  echo
31790/tcp  open  ssl/unknown
31960/tcp  open  echo
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.c
gi?new-service :
SF-Port31790-TCP:V=7.94SVN%T=SSL%I=7%D=5/30%Time=6A1AFD34%P=x86_64-pc-linu
SF:x-gnu%r(GenericLines,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x2
SF:0current\x20password\.\n")%r(GetRequest,32,"Wrong!\x20Please\x20enter\x
SF:20the\x20correct\x20current\x20password\.\n")%r(HTTPOptions,32,"Wrong!\
SF:x20Please\x20enter\x20the\x20correct\x20current\x20password\.\n")%r(RTS
SF:RRequest,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x20current\x20
SF:password\.\n")%r(Help,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x
SF:20current\x20password\.\n")%r(FourOhFourRequest,32,"Wrong!\x20Please\x2
SF:0enter\x20the\x20correct\x20current\x20password\.\n")%r(LPDString,32,"W
SF:rong!\x20Please\x20enter\x20the\x20correct\x20current\x20password\.\n")
SF:%r(SIPOptions,32,"Wrong!\x20Please\x20enter\x20the\x20correct\x20curren
SF:t\x20password\.\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 162.68 seconds
```

31790포트에 SSL/TLS 암호화를 사용해서 현재 비밀번호를 전달하니 RSA 개인키가 나왔다.

```
bandit16@bandit:~$ echo kSkvUpMQ7LBYyCM4GBPvCvT1BfWRy0Dx | openssl s_client -connect localhost:31790 -ign_eof -quiet
Can't use SSL_get_servername
depth=0 CN = SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN = SnakeOil
verify return:1
Correct!
-----BEGIN RSA PRIVATE KEY-----
MIIEogIBAAKCAQEAvM0kuiMfMg6HL2YPI0jon6iWfbp7c3jx34YkYWqUH57SudyJ
imZzeyGC0gt2PGuJUSxiJSWI/oTqexh+cAMTSMLOJf7+BrJ0bArnxd9Y7YT2bRPQ
Ja6Lzb558YW3FZL870Ri0+rWuLDCDCNd2LuvLE/GL2GWyuKN0K5iCd5TbtJzEkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WwAW18BR6yGrMq7Q/kALHYW30ekePQAZL0VUYbW
JGTi65CxbCnzc/w4+mqQyvmzpwMAzJTzAzQxNbkR2MBGySxDLrjg0LWN6sk7wNX
x0YVztz/zBikPjfkU1jHS+9EbVNj+D1XF0JuaQIDAQABAoIBABagpXPm1aolWfvD
KHcj10nqcoBc4oE1laFYQwik7xfW+24pRNUDE6SFth0ar69jps5RLwD1NhPx3iBl
J9n0M80J0VToum43UOS8YxF8WwhXriYGnc1sskbwpX0UDc9uX4+UESzH22P29ovd
d8WErY0gPxn8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9ALbssgTcCXkMQnPw9nC
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52y0Q9q0kwFTEQjpTf4uNtJom+asvLpmS8A
vLY9r60wYSvmZhNgBURj7lyCtXMIu1kkd4w7F77k+DjHoAXyXUp1DGL51sOmama
+TOWWgECGYEA8JtPxP0GRJ+IQkX262jM3dEIkza8ky5moIwUqYdsx0NxHGRRH0RT
8c8hAuRBb2G82so8vUHK/fur850Efc9TncnCY2crpqsgshifKLxrLgtT+qDpFZnx
SatLdt8GfQ85yA7hnWWJ2Mx3NaesDm75Lsm+tBbAiyc9P2jGRNtMSkCgYEAYpHd
HCctNi/FwjuLhtFf/rHYKHLidZDFYeIE/v45bN4yFm8x7R/b0iE7KaszX+Exdvt
SghaTdcG0Knyw1bpJVYusavPzpaJMjdJ6tcFhVABAjm7enCivGCSx+X3L5SiWg0A
R57hJgLeziIvJv3aGwHwvLZvtszK6zV6oXFAu0ECGYAbjo46T4hyP5tJi93V5Hdi
TtieK7xRVxU+iu7rWkGAXFpMLFteQEsRr7PJ/LemMEY5eTDAFmLy9FL2m9oQWCg
R8VdWsk8r9FGLS+9aKcV5PI/WEKlwgXinB30hYimtI62Cg5JCqIZFHx0D6MjEG0iu
L8ktHMPvodBwNsSBULpG0QKBgBAPLtfC1H0nWiMGOU3KPwYwT006CdTkmJ0mL8Ni
blh9elyZ9FsGxsgtRBXRsqXuz7wtsQAgLHxbdLq/ZJQ7Yfz0KU4ZxEnabvXnvWkU
Y0djHdS0oKvDQNWu6ucyLRAWFuISeXw9a/9p7f7tpxm0TSgyvmfLF2MIAEwyzRqaM
77pBAoGAMmjmiJdjp+Ez8duyn3ieo36yrftF5NSsJLABxPdpLc1gvtGCWW+9Cq0b
dxviW8+TFVEBL104f7HVM6EpTscdXu+bCXWkfjuRb7Dy9G0tt9JP5X8MBTakh3
vBgysi/sN3RqRBcGU40f0oZyFAMT8s1m/uYv5206IgeuZ/ujbjY=
-----END RSA PRIVATE KEY-----
```

nano 명령어로 key.private을 만들어 위에서 구한 RSA 개인키를 구하고 권한을 부여했다.

```
user@DESKTOP-CS2C1DH:~$ nano key.private
user@DESKTOP-CS2C1DH:~$ chmod 600 key.private
```

bandit17에 key.private을 이용해 접속하고 cat으로 경로를 찾으니 비밀번호가 나왔다.

```
bandit17@bandit:~$ cat /etc/bandit_pass/bandit17
EReVavePLFHtFlFsijn3hyzMlvSuSAcRD
```

Level 17 → Level 18

비밀번호는 두 파일 passwords.old와 passwords.new의 차이점 중 passwords.new에 들어있는 내용이라고 한다. diff 명령어로 차이점을 파악한다. 둘 중 passwords.new에 들어있는 내용은 > 뒤의 내용이다.

```
bandit17@bandit:~$ ls
passwords.new  passwords.old
bandit17@bandit:~$ diff passwords.old passwords.new
42c42
< 390zFj2NETFVZkqYw8UEFdN6h40oGVtT
---
> x2gLTTjFwM0hQ8oWNbMN362QKxfRqGL0
```

Level 18 → Level 19

비밀번호는 readme 파일에 있는데 SSH로 로그인은 안된다고 한다. 실제로 평범하게 접속하니 바로 연결이 끊겼다.

```
Byebye !
Connection to bandit.labs.overthewire.org closed.
```

SSH 접속에 cat readme 명령어를 붙여 실행하니 비밀번호를 얻을 수 있었다.

```
user@DESKTOP-CS2C1DH:~$ ssh bandit18@bandit.labs.overthewire.org -p 2220 cat readme

      _ _ _ _ _
     |   _   |
     |  (X)  |
     | _   _ |
     |  _   |
     | |   |
     |__|_||

      This is an OverTheWire game server.
      More information on http://www.overthewire.org/wargames

backend: gibbon-0
bandit18@bandit.labs.overthewire.org's password:
cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8
```

Level 19 → Level 20

파일 목록을 확인하니 bandit20-do 이름으로 setuid가 있다. 확인하니 사용법도 알려준다. 알려준대로 bandit20의 권한으로 비밀번호를 알 수 있었다.

```
bandit19@bandit:~$ ls -la
total 36
drwxr-xr-x  2 root    root    4096 Apr  3 15:17 .
drwxr-xr-x 150 root    root    4096 Apr  3 15:20 ..
-rwsr-x---  1 bandit20 bandit19 14888 Apr  3 15:17 bandit20-do
-rw-r--r--  1 root    root      220 Mar 31 2024 .bash_logout
-rw-r--r--  1 root    root    3851 Apr  3 15:10 .bashrc
-rw-r--r--  1 root    root     807 Mar 31 2024 .profile
bandit19@bandit:~$ ./bandit20-do
Run a command as another user.
Example: ./bandit20-do whoami
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
0qXahG8Zj0VMN9Ghs7i0WsCfZyX0UbY0
```